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Anand Nagar, Krishnankoil, Srivilliputtur (Via), Virudhunagar (Dt) - 626126, Tamil Nadu | info@kalasalingam.ac.in | www.kalasalingam.ac.in



WEB BASED E-WASTE AND BATTERY LIFECYCLE MANAGEMENT SYSTEM

A COURSE LEVEL PROJECT REPORT

Submitted by

III-year students of Bachelor of Technology

M.MIRUDHULA DAISY - 9923005207

S.SIVA PRIYA - 9923005224

S.DEEPADHARSHINI - 9923005245

AKUETI BUNNY - 99230040004

R.SIDDARTH - 9923005265

in partial fulfillment of the course of

215EXS3201 / EXSEL – DESIGN-BUILD-OPERATE

Academic Year 2025 – 2026 (Even Semester)

DECLARATION

We affirm that the project work titled “**WEB BASED E-WASTE AND BATTERY LIFECYCLE MANAGEMENT SYSTEM**” being submitted in partial fulfilment for the award of the degree of **Bachelor of Technology** is the original work carried out by us. It has not formed part of any other project work submitted for the award of any degree or diploma, either in this or any other University.

Reg. Number	Student Name	Student Signature
9923005207	M.MIRUDHULA DAISY	
9923005224	S.SIVA PRIYA	
9923005245	S.DEEPADHARSHINI	
9923005265	R.SIDDARTH	
99230040004	AKUETI BUNNY	

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Date:

Signature of supervisor

Mrs.S.Shanmugapriya

Assistant Professor

Department of Biomedical Engineering



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BONAFIDE CERTIFICATE

Certified that this project report **“WEB BASED E-WASTE AND BATTERY LIFECYCLE MANAGEMENT SYSTEM”** is the bonafide work of **“M.MIRUDHULA DAISY(9923005207),S.SIVAPRIYA(9923005224), S.DEEPADHARSHINI(9923005245),R.SIDDARTH(9923005265),AKUETI BUNNY(99230040004)”** who carried out the project work under my supervision.

Signature of supervisor

Mrs.S.Shanmugapriya

Assistant Professor

Department of Biomedical Engineering

Submitted for the Project Viva-voce / Review held at Kalasalingam Academy of Research and Education, Krishnankoil on

Internal Examiner

External Examiner

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PROJECT SUMMARY

Project Title	WEB BASED E-WASTE AND BATTERY LIFECYCLE MANAGEMENT SYSTEM	
Project Team Members (Name with Register No)	M.MIRUDHULA DAISY - 9923005207 S.SIVA PRIYA - 9923005224 S.DEEPADHARSHINI - 9923005245 AKUETI BUNNY - 99230040004 R.SIDDARTH - 9923005265	
Guide Name/Designation	Mrs.S.Shanmugapriya	
Program Concentration Area	Web-based e-waste and battery monitoring	
Technical Requirements	HTML, CSS, Python, MySQL	
Engineering standards and realistic constraints in these areas		
Area	Codes & Standards / Realistic Constraints	Tick ✓
Economic	Low maintenance cost	✓
Environmental	Safe e-waste disposal	✓
Social	Community awareness	✓
Ethical	Secure data handling	✓
Health and Safety	Prevents battery hazards	✓
Manufacturability	Easy web deployment	✓
Sustainability	Supports recycling and SDG 12	✓

ABSTRACT

The rapid growth of electronic devices in educational institutions has led to an increase in electronic waste generation. Devices such as computers, printers, and UPS systems depend on batteries whose performance reduces over time. In many universities, maintenance activities are still handled manually, making it difficult to monitor battery condition and schedule timely replacement. This paper introduces a web-based E-Waste and Battery Lifecycle Management System that helps institutions register devices, record maintenance details, assess battery condition, and generate alerts for replacement. The system also provides a dashboard for analyzing device performance and e-waste trends. This approach improves maintenance. This work mainly supports SDG 12 (Responsible Consumption and Production) by promoting sustainable e-waste disposal and battery lifecycle management, and it also contributes to SDG 9 (Industry, Innovation and Infrastructure) through an innovative web-based monitoring platform.

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LIST OF ABBREVIATION

1. **UPS** -Uninterruptible Power Supply
2. **IP** -Internet Protocol
3. **ID** -Identification
4. **HTML** – HyperText Markup Language
5. **CSS** – Cascading Style Sheets
6. **JS** – JavaScript
7. **PHP** – Hypertext Preprocessor
8. **SQL** – Structured Query Language
9. **DB** – Database
10. **OS** – Operating System
11. **UI** – User Interface
12. **CPU** – Central Processing Unit
13. **RAM** – Random Access Memory
14. **LAN** – Local Area Network

PROBLEM STATEMENT

The continuous use of electronic devices such as computers, UPS systems, printers, routers, and laboratory instruments in educational institutions has led to a rapid increase in battery degradation and electronic waste generation. As these devices age, their batteries lose efficiency due to repeated charging cycles, temperature variations, and prolonged operational use, which may result in unexpected failures and service interruptions.

In many institutions, maintenance activities are still managed using manual registers, spreadsheets, or paper-based logs. These traditional approaches make it difficult to maintain accurate records of battery health, replacement history, and disposal details, especially when devices are distributed across multiple departments and laboratories. As a result, early signs of battery failure are often missed, leading to delayed maintenance and higher replacement costs.

In addition, improper handling of expired batteries and discarded electronic equipment creates environmental risks because such waste contains hazardous materials that require safe disposal and recycling. The absence of a centralized monitoring and lifecycle tracking system further increases the chances of poor e-waste management.

Therefore, there is a strong need for a web-based E-Waste and Battery Lifecycle Management System that can provide centralized device registration, battery condition monitoring, alert generation, maintenance history tracking, and sustainable disposal support.

CHAPTER –I

INTRODUCTION

Electronic devices are widely used in universities for academic, research, and administrative purposes. Systems such as computers, printers, networking equipment, and UPS units are essential for daily operations. As the number of these devices increases, the amount of electronic waste generated also rises.

Among all hardware components, batteries require special attention because their efficiency decreases with repeated charging cycles, temperature changes, and continuous use. If battery health is not checked regularly, unexpected device failures may occur, affecting institutional activities.

Many institutions still depend on manual logs or spreadsheet-based maintenance records. This makes it difficult to track device condition, identify battery degradation, and plan preventive maintenance. As a result, device failures are often detected only after they occur.

To solve this problem, the proposed system offers a centralized web-based platform for monitoring devices, evaluating battery health, and generating alerts. The system also supports lifecycle tracking and e-waste analysis, helping institutions improve maintenance planning and sustainable device usage.

CHAPTER-II

LITERATURE REVIEW

Recent studies show the increasing role of digital technologies in electronic waste management and device lifecycle supervision. Early research by Shiby and Wijegoda [1] highlighted the importance of web-based platforms in organizing e-waste records and improving recycling efficiency through centralized data systems. In a similar direction, Armstrong et al. [2] focused on monitoring the movement of e-waste from collection centers to recycling facilities, thereby improving transparency in disposal workflows.

Systematic approaches for e-waste handling were further discussed by Suma [3] and Spoorthi [4], who proposed web-based solutions for efficient waste data storage, tracking, and reporting. Although these systems improved digital management, they lacked advanced monitoring and predictive capabilities. To address real-time challenges, Kumar et al. [5] introduced an IoT-based waste monitoring framework capable of observing waste levels dynamically, though it required additional sensing hardware.

Independent studies have also explored device maintenance and health supervision. Ravi et al. [6] proposed a maintenance logging system with failure prediction support, while Patel et al. [7] developed a cloud-enabled platform for managing e-waste records across multiple locations. Singh et al. [8] emphasized rule-based battery health evaluation using parameters such as voltage, temperature, and cycle count.

The concept of lifecycle tracking and sustainability analytics was extended by Rahman et al. [9], who proposed usage-based replacement estimation methods. Similarly, Chen et al. [10] applied data-driven techniques to study e-waste generation patterns and support maintenance planning. IoT-enabled battery monitoring solutions such as Li et al. [11] provided accurate real-time battery status, but the increased hardware dependency raised system complexity and deployment cost.

Advanced predictive and intelligent maintenance techniques have also been investigated. Zhang et al. [12] used machine learning methods for forecasting device failures, while Ahmed et al. [13] improved asset traceability using web-based lifecycle supervision. Verma et al. [14] and Khan et al. [22] introduced alert-driven maintenance frameworks that notify administrators about possible failures, thereby supporting preventive actions.

Recently, integrated monitoring and analytics systems have received greater attention. Joseph et al. [15] and Iyer et al. [23] proposed multi-functional frameworks combining tracking, reporting, and health monitoring. Nair et al. [16] and Sharma et al. [24] developed dashboard-oriented systems for visualizing device performance and e-waste trends. In addition, Gupta et al. [17] and Mehta et al. [21] highlighted centralized cloud-based architectures for scalable and accessible institutional monitoring.

Other notable contributions include embedded and statistical techniques for battery fault detection. Hassan et al. [18] implemented embedded real-time monitoring solutions, while Roy et al. [19] applied statistical prediction models for identifying battery failure trends. Lightweight and economical frameworks were later introduced by Das et al. [20], making such systems suitable for smaller institutions.

From the reviewed literature, it is evident that most existing works mainly focus on individual aspects such as waste tracking, battery monitoring, predictive maintenance, or dashboard reporting. However, only limited research combines device registration, battery health evaluation, lifecycle tracking, alert generation, and analytical visualization into a single integrated platform. Therefore, the proposed work [25] bridges this gap by developing a comprehensive Web-Based E-Waste and Battery Lifecycle Management System.

CHAPTER-III

PROBLEM DEFINITION AND PROJECT OBJECTIVES

The increasing use of computers, UPS systems, printers, networking devices, and laboratory equipment in educational institutions has resulted in a significant rise in battery degradation and electronic waste generation. Over time, batteries lose efficiency due to repeated charging cycles, temperature variations, and continuous operation, which may lead to sudden device failures and service interruptions.

At present, many institutions still rely on manual maintenance records, spreadsheets, or paper-based logs to monitor device condition and replacement history. These traditional methods make it difficult to maintain accurate lifecycle data, especially when devices are distributed across multiple departments and laboratories.

As a result, battery failures are often detected only after the device stops functioning, leading to delayed maintenance and higher replacement costs. In addition, the absence of a centralized monitoring system makes it challenging to plan safe disposal, reuse, or recycling of expired batteries and outdated devices. This increases environmental risks and contributes to improper e-waste management.

Hence, there is a need for a web-based centralized solution that can track device details, monitor battery health, generate timely alerts, maintain lifecycle records, and support sustainable e-waste handling.

Project Objectives

The main objectives of the proposed project are:

- To develop a web-based platform for centralized monitoring of institutional electronic devices.
- To register and track device details such as location, installation date, and battery specifications.
- To evaluate battery health condition using parameters like voltage, temperature, and charging cycles.
- To generate timely alerts for battery replacement and preventive maintenance.
- To maintain lifecycle history records for devices and batteries.
- To provide dashboard-based analytics for device performance and e-waste generation trends.
- To support sustainable disposal and recycling decisions for expired batteries and outdated devices.

CHAPTER-IV

PROPOSED METHODOLOGY

The proposed system follows a structured methodology to monitor electronic devices and manage battery lifecycle within a university environment. The methodology consists of several sequential stages including device registration, periodic maintenance data entry, battery evaluation, alert generation, and analytical reporting. Each stage ensures efficient monitoring and proper management of electronic waste.

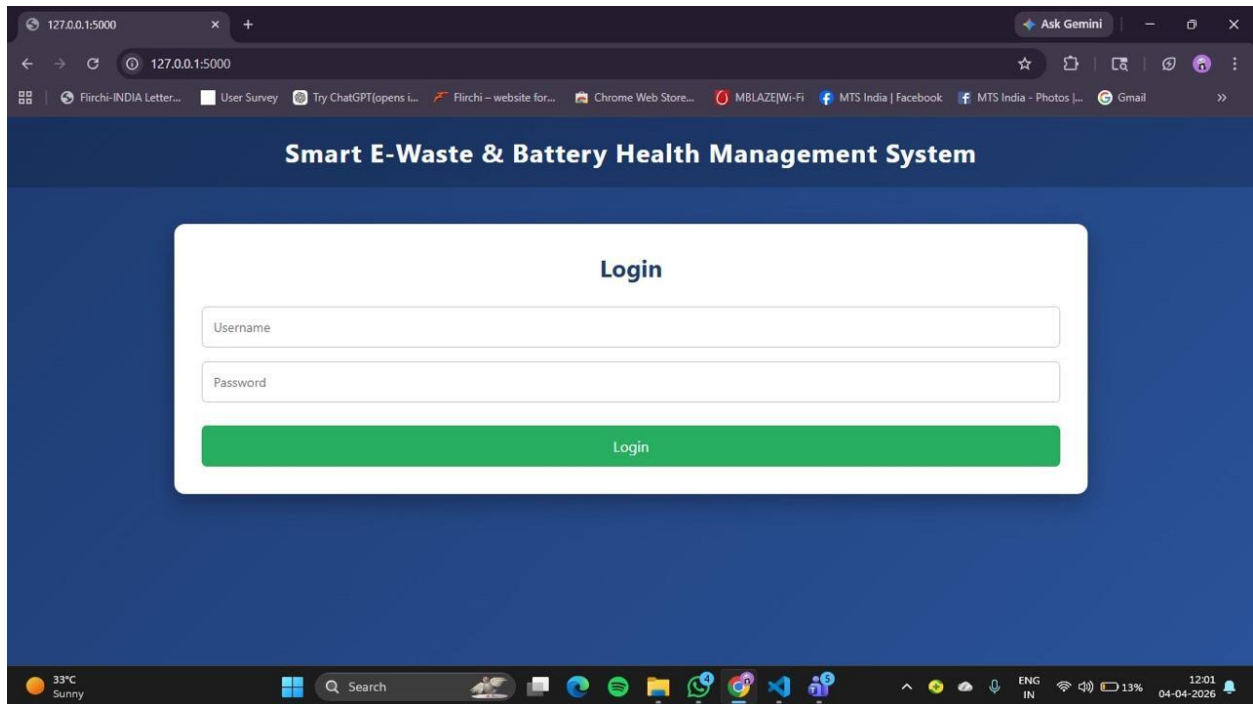


Fig 1:Admin page

4.1 Location Identification:

The fig 2 of the system involves identifying and registering all operational areas within the institution. These locations include computer laboratories, research laboratories, library systems, academic departments, administrative offices, and IT maintenance rooms. The administrator performs this process during the initial configuration of the system. Defining these locations helps create an organized structure for managing devices and tracking their operational status across different areas of the institution.

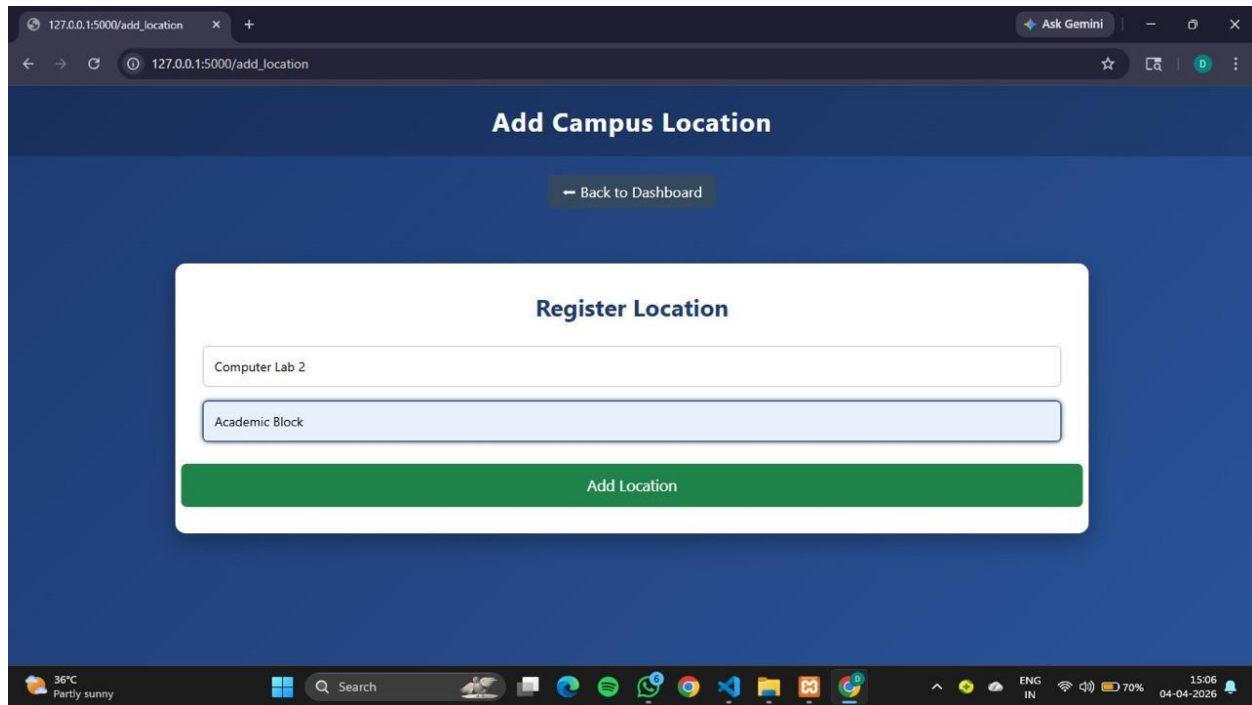


Fig 2:Location page

4.2 Device Registration Page

After defining the locations, in fig 3 all electronic devices used in the institution are registered in the system. Each device entry contains important details such as device type, installation date, battery specifications, and physical location. Once the device is registered, the system generates a unique digital identification number. This digital identity allows the system to track the device throughout its lifecycle and maintain a history of maintenance activities and performance records.

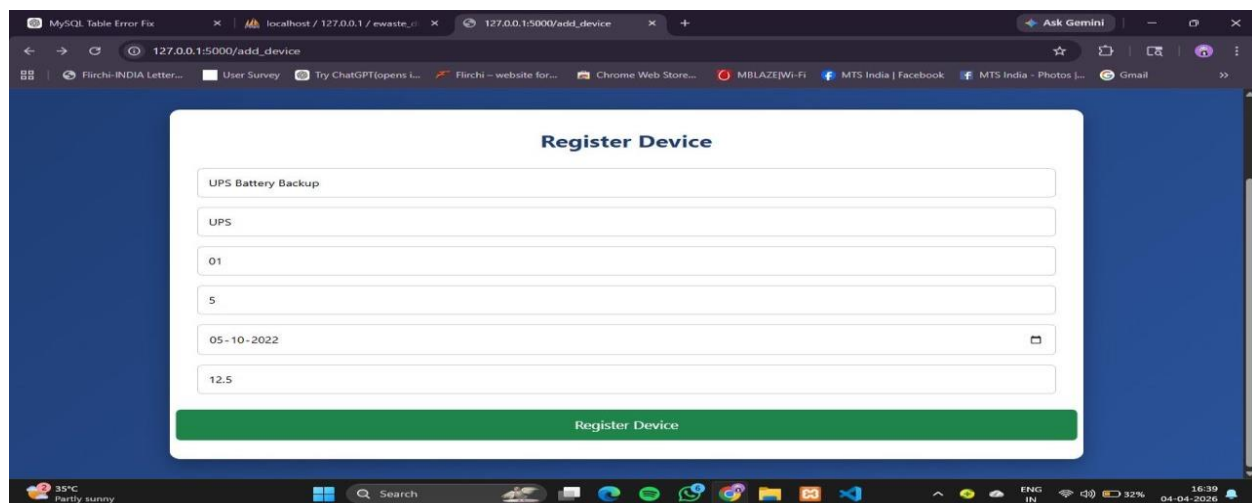


Fig 3:Device Registration Page

4.3 Periodic Maintenance Data Entry:

At regular intervals such as weekly or monthly, IT maintenance staff enter device parameters into the system. These parameters include battery voltage level, operating temperature, and charge cycle information. The system relies on structured manual inputs instead of additional hardware sensors. This approach reduces implementation cost while still allowing the system to monitor device performance effectively using existing maintenance records are involved in the fig 4

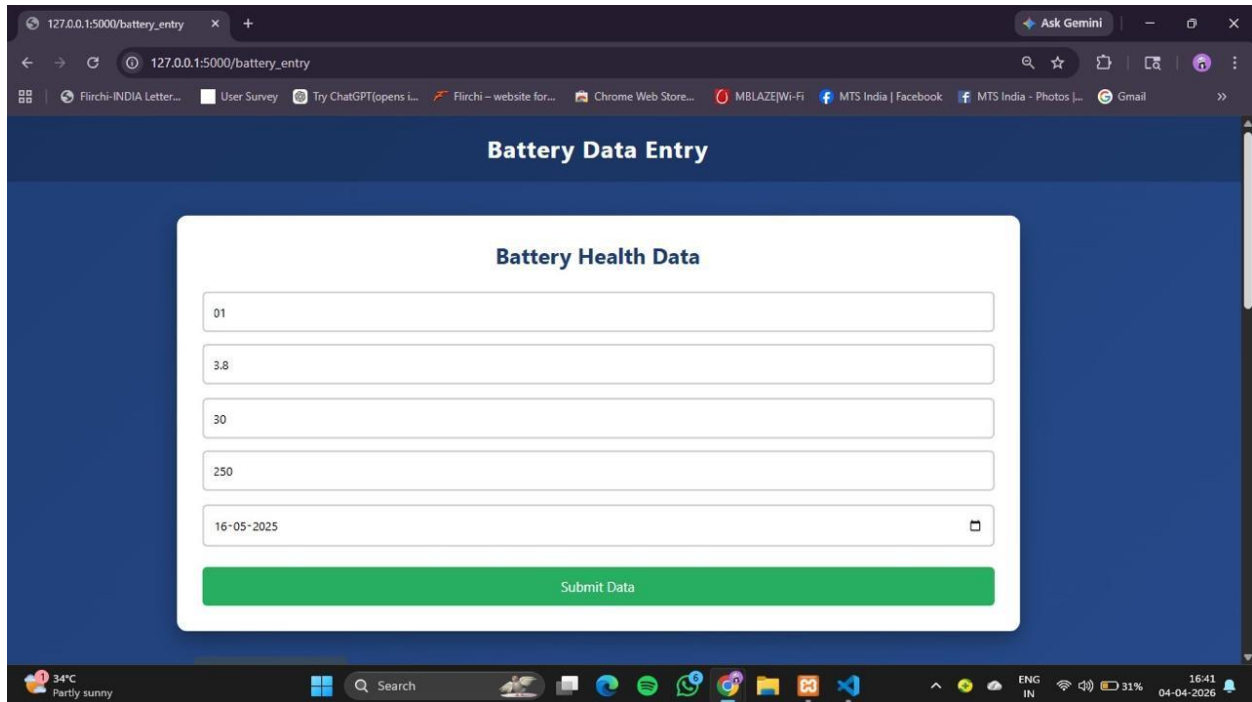
The image shows a web browser window displaying a web application titled "Battery Data Entry". The application has a dark blue header with the title. Below the header is a white form titled "Battery Health Data". The form contains five input fields: a text field with "01", a text field with "3.8", a text field with "30", a text field with "250", and a date field with "16-05-2025". Below these fields is a green button labeled "Submit Data". The browser's address bar shows the URL "127.0.0.1:5000/battery_entry". The Windows taskbar is visible at the bottom, showing the date and time as "16:41 04-04-2026".

Fig 4: Periodic Maintenance Data Entry

4.4 Battery Condition Evaluation:

After maintenance data is entered, in fig 5 the system evaluates the condition of each battery using predefined rule-based calculations. The evaluation process analyzes the recorded parameters and estimates the overall health status of the battery. Based on the calculated values, the system classifies the device condition into different operational states such as normal condition, monitoring stage, replacement requirement, or disposal stage. This classification helps administrators identify devices that require immediate attention.

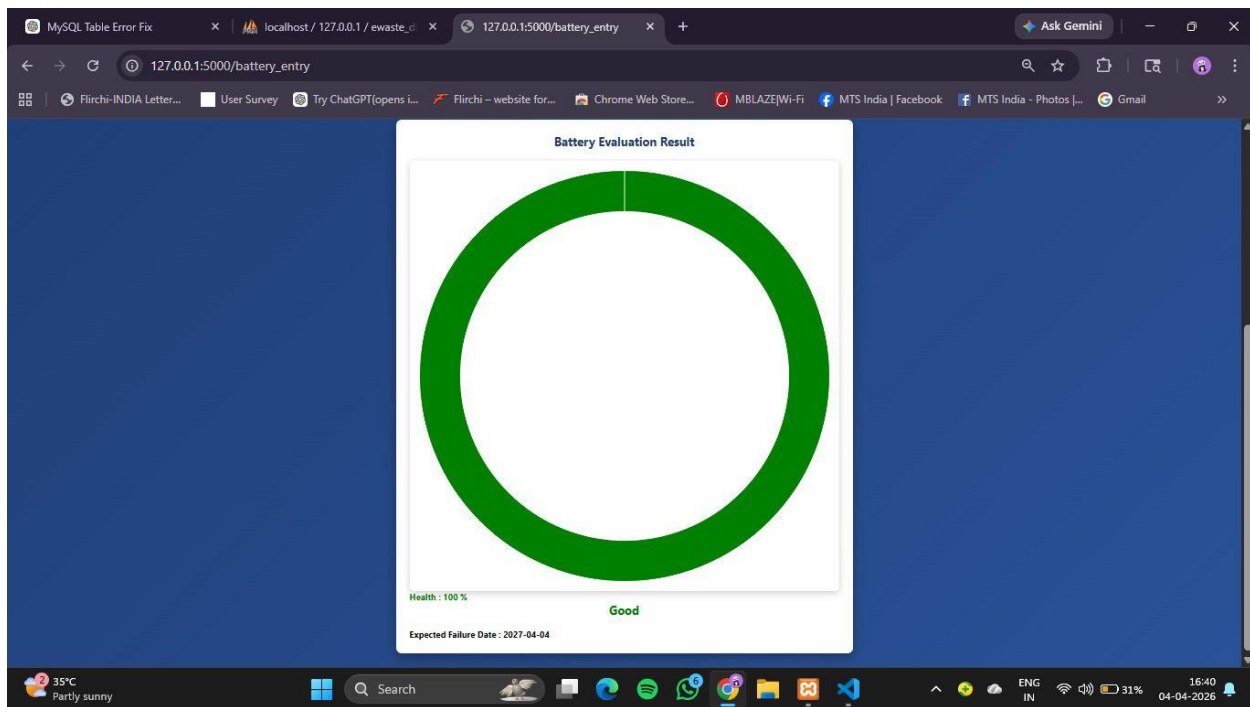


Fig 5:Battery Condition Evaluation:

4.5 Data Storage and Lifecycle Tracking:

All device information, maintenance records, and evaluation results are stored in a centralized database. The database maintains historical data for each registered device, enabling long-term monitoring and lifecycle tracking. This stored information also assists administrators in identifying usage patterns, maintenance frequency, and device replacement requirements. In fig 6, fig 7 are used for the data entry storage evaluated.

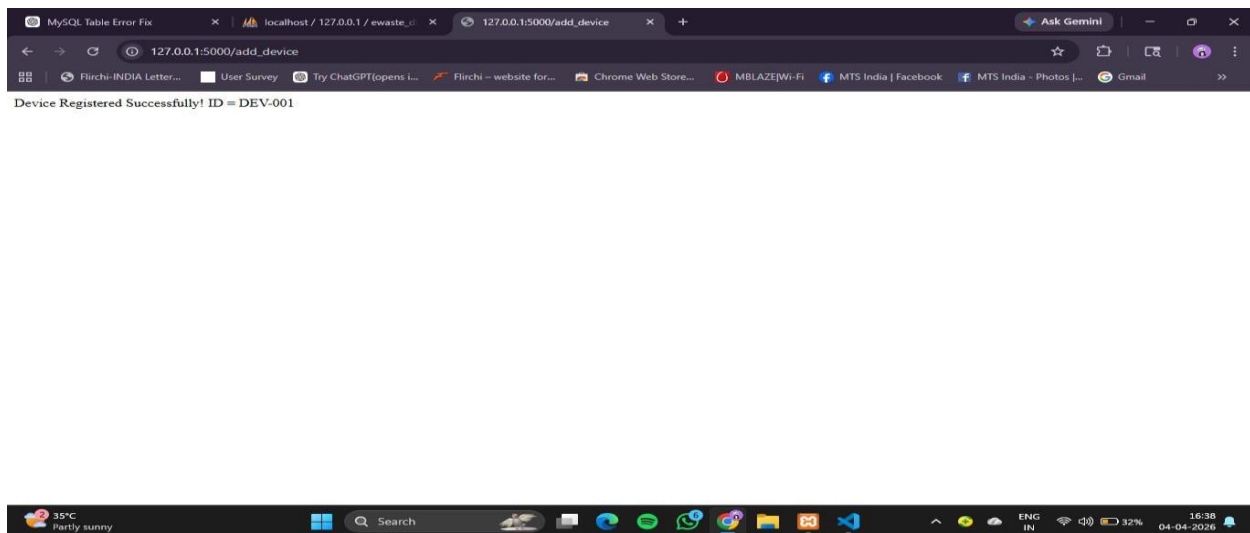


Fig 6:Device storage

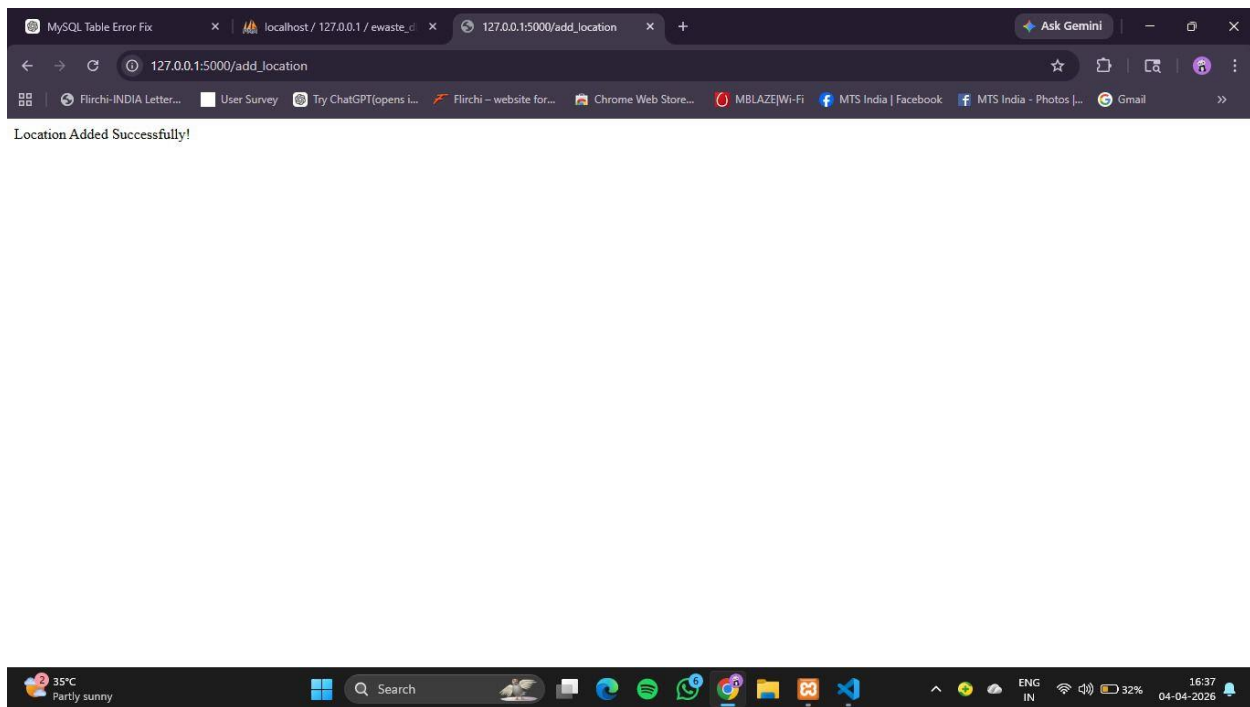


Fig 7:Data Storage

4.6 Analytics and Report Generation:

The final stage of the fig 8 involves generating analytical reports based on stored data. The system automatically produces graphical charts and summaries that display battery replacement trends, device performance statistics, and electronic waste generation patterns. These reports assist administrators in planning maintenance activities, managing device inventory, and implementing environmentally responsible disposal.

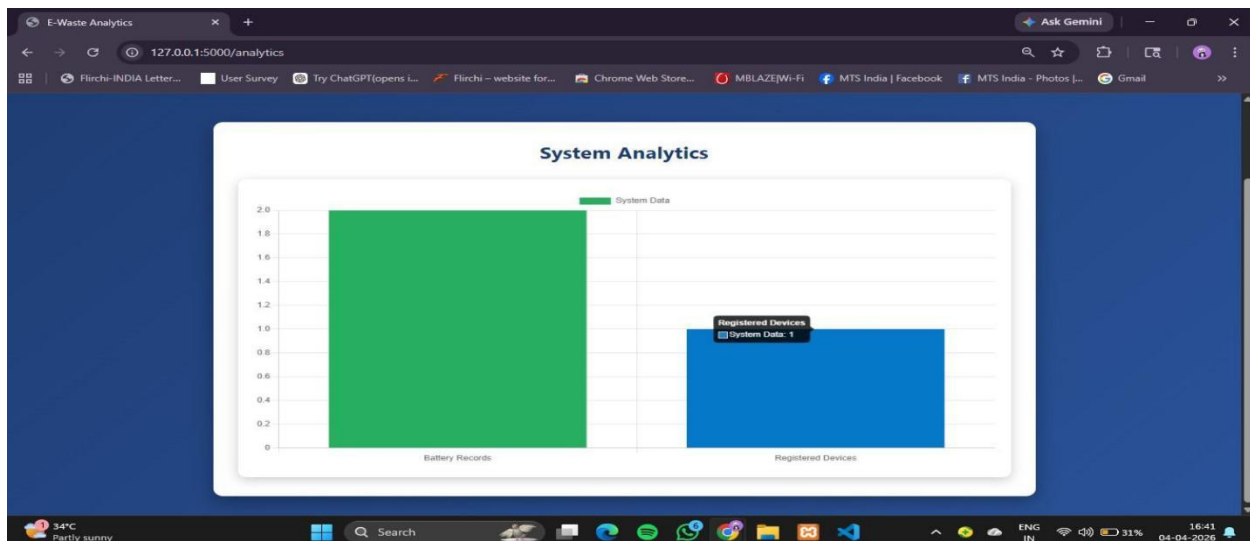


Fig 8:Analytics

CHAPTER-V

REQUIREMENTS AND MODULE DESCRIPTION

The successful implementation of the proposed Web-Based E-Waste and Battery Lifecycle Management System requires both hardware and software resources.

5.1 Software Requirements

- Frontend: HTML, CSS, JavaScript
- Backend: Python / PHP / Node.js
- Database: MySQL / MongoDB
- Framework: Django / Flask / React (optional)
- Operating System: Windows / Linux
- Browser: Google Chrome / Microsoft Edge

Module Descriptions

The proposed system is divided into the following functional modules:

User Authentication Module

This module provides secure login access for administrators and maintenance staff. It ensures that only authorized users can update device records, battery details, and maintenance history.

Device Registration Module

This module is used to register all institutional electronic devices such as computers, UPS systems, printers, and routers. Details like device ID, location, installation date, battery type, and department are stored.

Battery Monitoring Module

The battery monitoring module records important battery parameters such as voltage, temperature, and charging cycle count. These values are used to evaluate the health condition of the battery.

Health Evaluation Module

This module applies rule-based threshold logic to classify batteries into:

- Healthy
- Observe
- Replace Soon
- Dispose

It helps in identifying early battery degradation.

Alert Generation Module

When abnormal battery conditions are detected, this module automatically sends notifications or alerts for timely maintenance and replacement.

Lifecycle Tracking Module

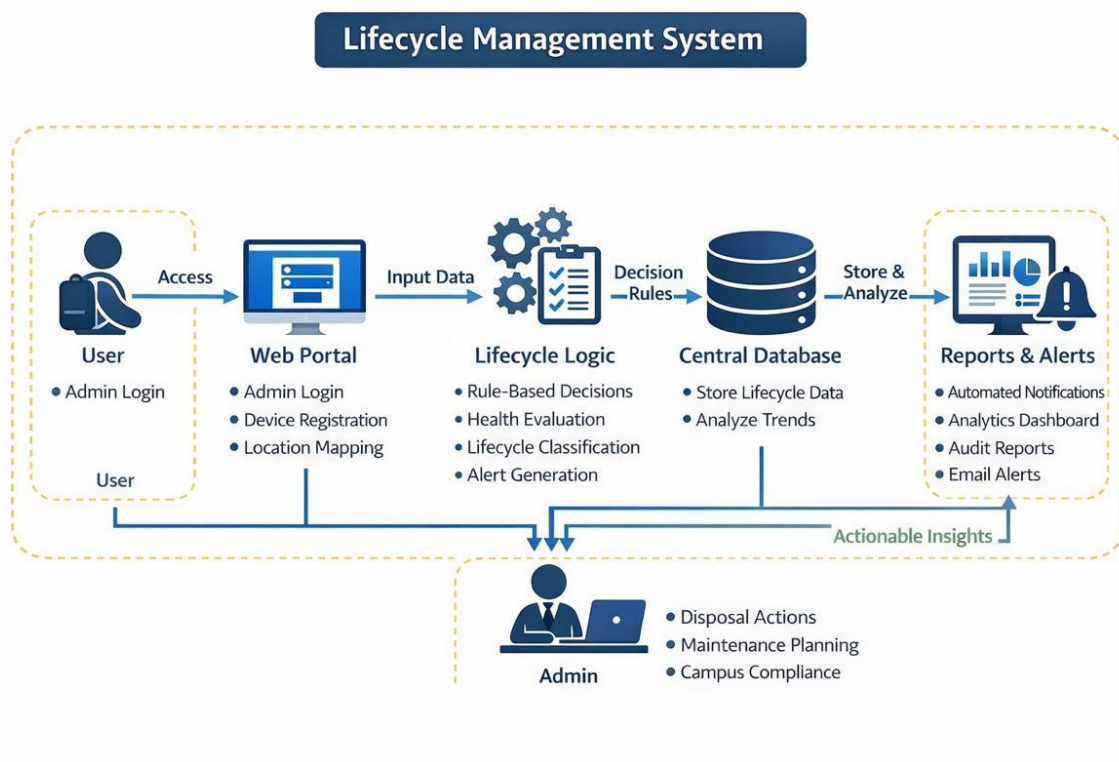
This module stores the complete maintenance and replacement history of each device. It supports tracking of battery lifespan, service frequency, and disposal stage.

Dashboard and Analytics Module

This module displays:

- battery degradation trends
- replacement frequency
- department-wise e-waste generation
- device performance reports

It helps administrators make better maintenance decisions.



Fg 9: System Module

CHAPTER-VI

SYSTEM IMPLEMENTATION

The implementation of the Web-Based E-Waste and Battery Lifecycle Management System was carried out using a modular web application architecture. The system was developed with a frontend interface for user interaction, a backend processing layer for logic execution, and a centralized database for lifecycle record storage.

Initially, the user authentication module was implemented to provide secure login access for administrators and maintenance staff. This ensured that only authorized users could register devices, update maintenance records, and access analytical reports.

The device registration module was then developed to store details of institutional electronic assets such as computers, UPS systems, printers, and networking devices. Each device was assigned a unique identification number along with attributes including department location, installation date, battery specifications, and maintenance schedule.

The battery monitoring module was implemented to periodically collect parameters such as voltage level, temperature, and charging cycle count. These values were entered through the web interface and stored in the database for further evaluation.

A rule-based health evaluation engine was integrated to analyze the collected battery parameters. Threshold logic was used to classify battery status into Healthy, Observe, Replace Soon, or Dispose. Based on the evaluated condition, the alert generation module automatically triggered maintenance notifications for batteries nearing failure.

The lifecycle tracking module was implemented to maintain historical records of repairs, replacements, and disposal actions. This made it possible to analyze battery wear patterns and identify devices with frequent failures.

Finally, the dashboard and analytics module was developed to display graphical reports of:

- Battery degradation trends
- Replacement frequency
- Device health statistics
- Department-wise e-waste generation

The completed system was tested using sample institutional devices, and the implementation successfully demonstrated centralized device supervision, timely battery alerts, and improved e-waste lifecycle visibility.

CHAPTER-VII

RESULTS AND DISCUSSION

The proposed Smart E-Waste Monitoring and Battery Lifecycle Management System, as shown in Fig.9 and Fig.10, was successfully implemented to supervise institutional electronic devices and evaluate battery health conditions. The system was tested using multiple assets such as computers, UPS units, and printers located across laboratories and administrative office environments.

During the testing phase, maintenance parameters including battery voltage, operating temperature, and charging cycle count were periodically entered into the web platform. These values were processed by the rule-based evaluation module, which analyzed the battery condition using predefined threshold calculations.

Based on the evaluated parameters, the system effectively classified devices into multiple operational states such as normal condition, observation stage, and replacement requirement. This classification enabled maintenance administrators to identify weak batteries before complete failure occurred.

The analytics module further generated graphical outputs showing:

- Battery health variation trends
- Monthly device failure statistics
- Department-wise e-waste generation patterns

These visual reports made it easier for administrators to understand long-term device performance, analyze failure frequency, and identify equipment requiring immediate maintenance or replacement.

The obtained results clearly demonstrate that the proposed system improves device monitoring efficiency, battery lifecycle supervision, and preventive maintenance planning. By combining structured asset tracking with automated analytical reporting, the platform also strengthens sustainable institutional e-waste management practices.

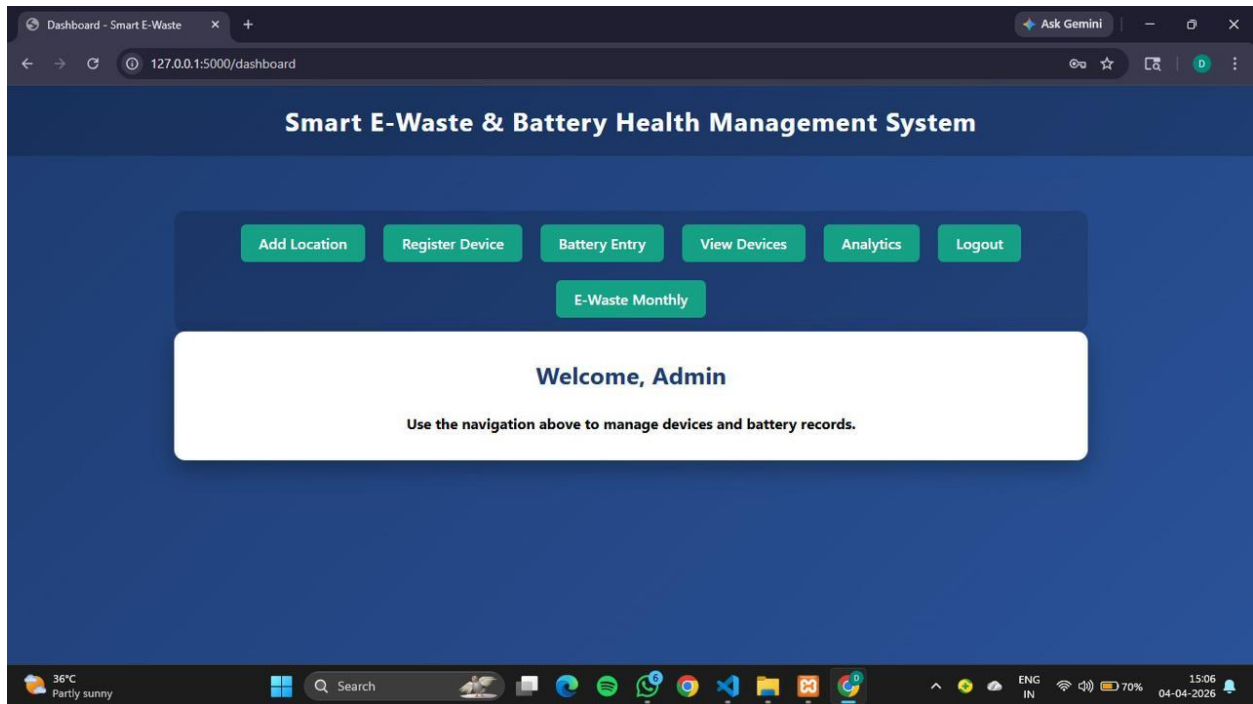


Fig 10:Dashboard

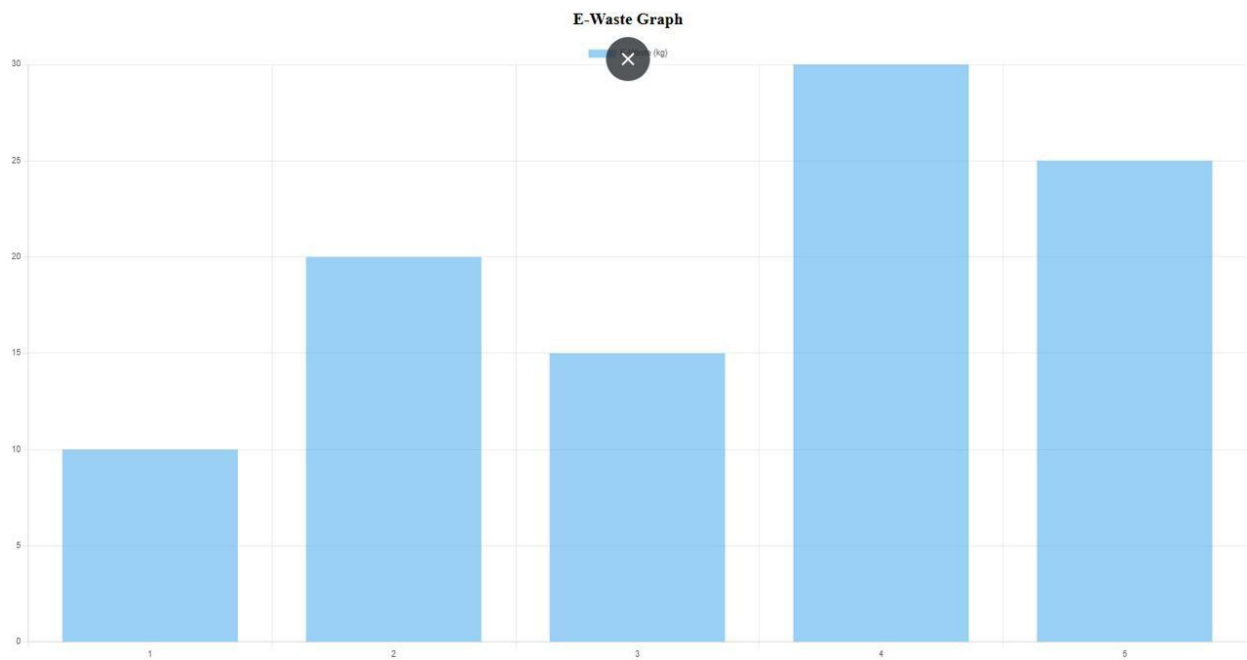


Fig 11:E-Waste generate

DISCUSSION

The proposed Web-Based E-Waste and Battery Lifecycle Management System offers significant improvements over existing e-waste and device monitoring solutions. Earlier systems mainly focused on individual functionalities, such as web-based waste tracking [108], cloud data storage [109], battery health evaluation [110], or IoT-based monitoring [111]. While these approaches addressed specific problems, they lacked a single integrated platform for complete device lifecycle supervision.

Parameter	Existing Systems	Proposed System
E-waste Tracking	Supports only waste record management	Includes waste tracking with lifecycle history
Battery Monitoring	Separate battery monitoring modules	Integrated battery health evaluation
Data Entry Method	IoT/manual depending on system	Low-cost structured manual web entry
Alert Generation	Available only in selected systems	Automatic battery replacement alerts
Lifecycle Tracking	Limited device replacement history	Full device and battery lifecycle tracking
Analytics Dashboard	Basic reporting or standalone graphs	Centralized dashboard with trends and reports

Cost	Higher for IoT and cloud systems	Cost-effective for institutions
Overall Integration	Individual functionalities	Complete all-in-one web
Sustainability Support	Focused mainly on disposal	Reuse, recycle, replacement, disposal support

Table 1: Comparison with Existing Systems

CHAPTER-VIII

COMMUNITY IMPACT

The proposed Web-Based E-Waste and Battery Lifecycle Management System creates a positive impact on the community by promoting safe electronic waste handling, efficient resource usage, and environmental sustainability. Educational institutions, offices, and laboratories generate a large number of outdated devices and used batteries, which, if not managed properly, may cause environmental pollution and health risks.

By providing centralized monitoring of device condition and battery lifecycle, the system helps institutions identify devices that can be reused, repaired, recycled, or safely disposed of. This reduces unnecessary dumping of electronic components and minimizes hazardous waste generation in the community.

The platform also improves preventive maintenance practices, which increases the lifespan of devices and reduces replacement costs. This benefits not only institutions but also society by encouraging responsible use of electronic resources.

In addition, the dashboard-based reporting system raises awareness about e-waste trends and battery disposal patterns, helping administrators and students understand the importance of sustainable technology usage.

Overall, the project contributes to a cleaner campus environment, reduced environmental pollution, better resource utilization, and stronger community awareness toward responsible e-waste management, thereby supporting SDG 12 and SDG 9.

CHAPTER-IX

CONCLUSION AND FUTURE SCOPE

The proposed Web-Based E-Waste and Battery Lifecycle Management System provides an effective solution for centralized monitoring of institutional electronic devices and battery conditions. By integrating device registration, maintenance record storage, battery health evaluation, alert generation, and dashboard-based analytics, the system improves preventive maintenance efficiency and reduces unexpected failures.

The implemented platform also supports better e-waste lifecycle tracking, enabling institutions to make informed decisions regarding battery replacement, reuse, recycling, and safe disposal. This contributes to sustainable resource utilization and environmentally responsible management of electronic assets.

FUTURE WORK

As a future scope, the system can be enhanced by integrating IoT sensors for automatic real-time collection of battery parameters such as voltage, temperature, and charging cycles. In addition, machine learning models can be incorporated to predict battery degradation and device failures at an earlier stage. Cloud deployment and mobile application support may further improve scalability and remote monitoring across multiple campuses or office locations.

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ENCLOSURES

1. Plagiarism Report



Page 2 of 9 - Integrity Overview

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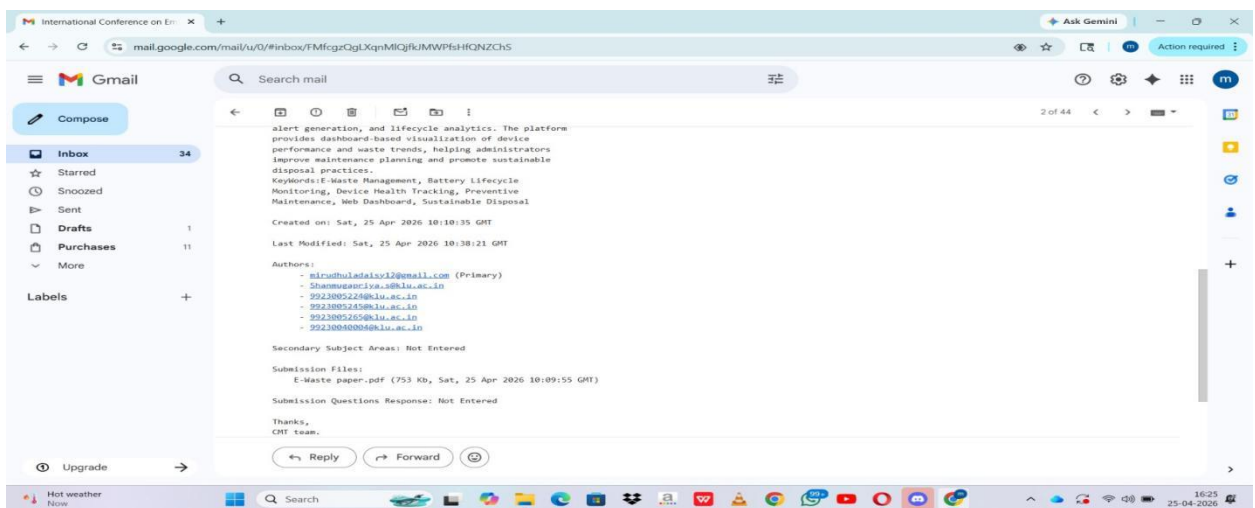
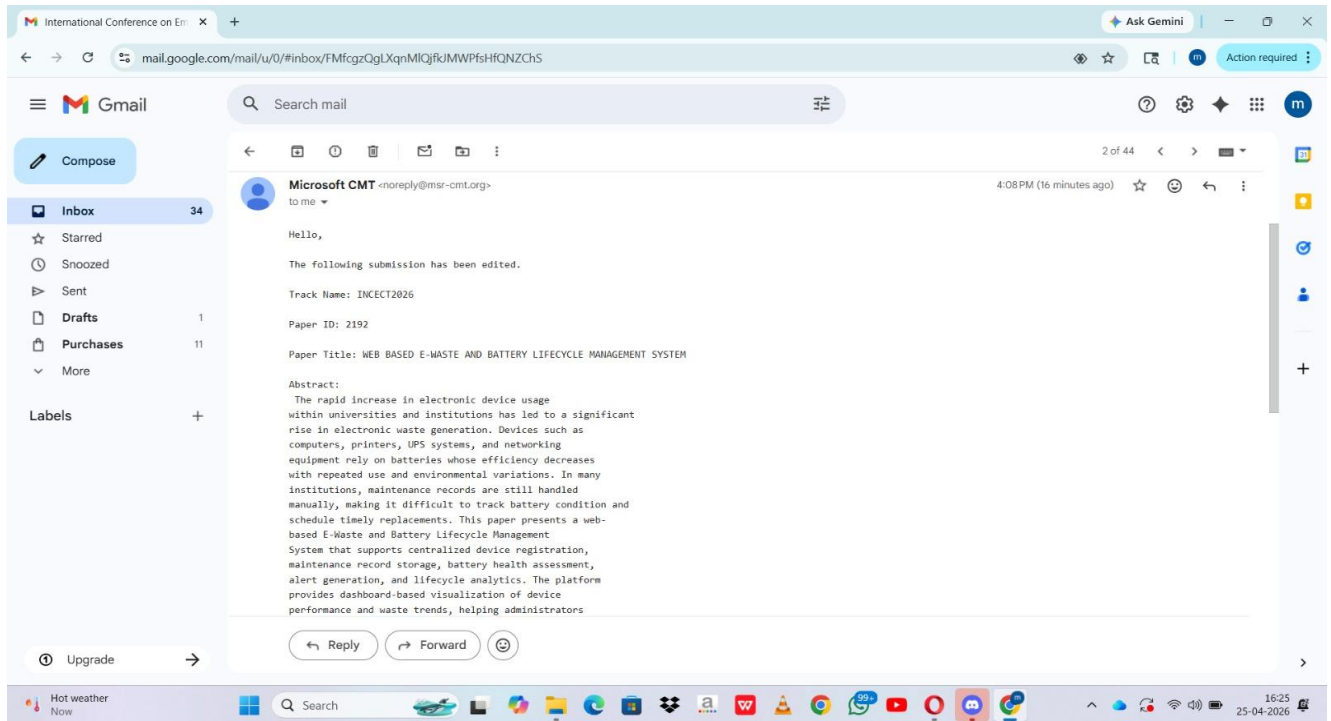
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2. Students Project related Achievements/Certificate/Outcome or photographs (If Applicable)

CERTIFICATE



PROOF OF SUBMISSION



3. Status of research Paper /Patent / research grant /Industry collaboration (3rd Year Project)

1) Research Paper

IEEE style research paper on “Web-Based E-Waste and Battery Lifecycle Management System” prepared and ready for conference/journal submission.

2) Patent

Concept-level innovation identified for centralized battery lifecycle monitoring and smart e-waste analytics; patent filing can be considered in future.

3) Industry Collaboration / Research Grant

Project suitable for collaboration with institutional IT departments, UPS maintenance vendors,



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INTERNAL QUALITY ASSURANCE CELL
PROJECT AUDIT REPORT

This is to certify that the project work entitled **“WEB BASED E-WASTE AND BATTERY LIFECYCLE MANAGEMENT SYSTEM”** categorized as an internal project done by **M.MIRUDHULA DAISY, S.SIVA PRIYA,S.DEEPADHARSHINI,AKUETI BUNNY and R.SIDDARTH** of the Department of Computer Science and Engineering, under the guidance of **Mrs.S.Shanmugapriya** during the Even semester of the academic year 2025 - 2026 are as per the quality guidelines specified by IQAC.

(Office use)

Quality Grade

Deputy Dean (IQAC)

Administrative Quality Assurance

Dean (IQAC)